

ECON 403/503—INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

Fall 2022

Department of Economics
Nazarbayev University

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Lecture time/Location: TR, 4:30 PM–5:45 PM, 7E220

Office hours: By appointment, book weekly at <https://alejandromelo.youcanbook.me>, please sign at least one day in advance.

Course Description This course analyzes problems and issues created by informational asymmetries and how properly designed market mechanisms could ameliorate those problems. Specifically, we will look at market mechanisms that can be classified as *auctions*. Auctions are currently used in a myriad of markets ranging from art to fresh fish, from electricity to government bonds, from government procurement to internet advertisements.

We will start the course by developing the tools required to analyze knowledge, beliefs, and information of economic agents. Then I will introduce the game theoretical model of *Bayesian Games*, which will be our main theoretical framework that we will use to model many auction formats.

After the introduction of the previous formal tools, we will examine the theoretical models behind the most commonly used auction formats. Students will learn the core theoretical results in auction theory.

Finally, we will introduce the basic theory of mechanism design in order to be able to discuss topics in auction design. Specifically, we will discuss the design of the optimal auction.

If time permits, I will discuss some advanced topics in auction theory, mechanism and information design.

Course Requirements ECON 201 Intermediate Microeconomics (C- and above) and ECON 301 Econometrics I (C- and above); however, these are just the *minimum, official requirements*. I will assume you are proficient with the topics studied in Calculus I, Probability, and Statistics, and that you are comfortable with analytical reasoning and mathematical notation in order to be able to follow the lectures and the readings. Therefore, it is strongly advised that you have a more extensive mathematical background than just a Calculus I course.

Furthermore, please note that this course has as an *unofficial prerequisite* knowledge of Game Theory, at the level of ECON302, or similar.¹ At the minimum, you should know what a normal-form game is, the concept of strategies, expected utility, dominance and rationalizability, Nash Equilibrium, mixed-strategy equilibrium, extensive-form games, and Subgame-Perfect Equilibrium.

Course Material There is no textbook required for this course. The course is based mainly on my lectures, slides and readings. All materials will be uploaded to Moodle.

The following is a list of undergraduate and graduate textbooks that you may find useful:

- Milgrom (2003), Putting Auction Theory to Work, Cambridge University Press.
- Krishna (2009), Auction Theory, Academic Press.

¹There is a popular course in Coursera with similar contents that could serve as a substitute for ECON302.

Grading Policy Grading will be weighted as follows:

For ECO403 (UG students)	For ECO503 (MA students)
Paper Report (12%)	Research Proposal (22%)
Problem Sets (20%)	Problem Sets (20%)
Midterm (30%)	Midterm (25%)
Final exam (35%)	Final exam (30%)
Attendance (3%)	Attendance (3%)

Problem Sets We will have at least 5 problem sets. They will be assigned in class and will be due the next week after they are assigned. Late submissions will **not** be accepted. I will grade your problem sets on a scale from 0 to 2 where:

- 0: Not delivered.
- 1: Delivered but incomplete.
- 2: Delivered and complete.

We will discuss solutions to the problem sets in class. *It is your responsibility to check your solutions against my class discussion to see if you were correct or where you made mistakes.* Problem sets are difficult and are meant to represent the level of difficulty that you will face in the exam problems.

Paper Report Undergraduate students have to read an academic paper and write a five-page report about it. I will provide more details about this assignment during the course.

Midterm The midterm will be on the third week of September. It will cover roughly half of the material in the course. The exact date will be announced during the course.

Research Proposal Master students have to work on a research proposal. The research proposal will have to be on a topic of your choice related to the general material of the course. You are welcome to talk to me to refine your topic. I will provide guidelines on how the research proposal should be written during the course.

Attendance All students are expected to attend the lectures and actively participate in class. If for some reason you are not able to attend to class, you should let me know in advance.

Final Exam Will cover the second half of the material. The date will be chosen by the Registrar, during Final's Week.

I will compute your final numeric grade by computing the weighted average of the aforementioned components according to the weights given in the table above. The numeric grade will then be curved to obtain the final letter grade.

Course attendance Attendance is compulsory. Students who are recorded as absent in 14 or more class meetings will receive an automatic F grade. This includes excused absences. For example, 4 excused absences and 10 unexcused absences count as 14 absences and lead to letter grade F. You will be recorded as absent if you are late by more than 10 minutes or leave early, use your phone, chat with your friends, do work for other classes, or engage in any kind of disruptive behavior. It is your responsibility to keep track of your attendance record.

Make-up Policy Students are expected to take all exams at the designated time. Make-up exams and homeworks will **not** be given. If you miss an exam or problem set due to serious illness, or injury of yourself or a close family member, its weight will be shifted to the remaining exams (problem sets) equally.

Rules and Regulations During lectures, please keep your cell phones switched off and refrain from text messaging. You are allowed to use your laptops or tablets to read the slides or take notes. Do not arrive late or leave early without prior permission. Do not engage in side conversations or other disruptive behaviours.

Academic Misconduct Academic misconduct is an act in which a student:

1. seeks to claim credit for the work or efforts of another without authorization or citation (“plagiarism”);
2. uses unauthorized materials or fabricated data in any academic exercise;
3. forges or falsifies academic documents or records;
4. intentionally impedes or damages the academic work of others;
5. engages in conduct aimed at making false representation of any student’s academic performance;
6. engages in any form of cheating on an exam or assignment;
7. assists other students in any of these acts

These are serious offenses subject to disciplinary action depending on their severity as detailed in the Student Code of Conduct.

Tentative Course Outline

- Knowledge and Beliefs.
- Games of Incomplete Information: Bayesian Games.
- Classical Auctions with complete information: First-price, Second-price, and English Auctions.
- Bayesian Auctions.
 - Private Value Auctions
 - Interdependent Value Auctions
- Revenue Equivalence and Linkage.
- Extensions: asymmetries, affiliation, risk averse bidders.
- Mechanism and Auction Design.
- Advanced Topics (time permitting).

Rules and Regulations During lectures, please keep your laptops, tablets and cell phones switched off and refrain from text messaging. Do not arrive late or leave early without prior permission. Do not engage in side conversations or other disruptive behaviors.

Academic Misconduct Plagiarism or cheating will not be tolerated and sanctioned according to the procedures in the Student Code of Conduct for Nazarbayev University. Students should familiarize themselves with this code. Please note that I do not discuss plagiarism cases unless a student makes an appeal to the disciplinary committee where my original penalty might be decreased or increased.

You can find the university’s official statement on plagiarism below:

Plagiarism is intentionally or carelessly presenting the work of another as one's own. It includes submitting an assignment purporting to be the student's original work which has wholly or in part been created by another person. It also includes the presentation of the work, ideas, representations, or words of another person without customary and proper acknowledgement of sources. Plagiarism occurs when a person:

- 1. Directly copies one or more sentences of another person's written work without proper citation. If another writer's words are used, you must place quotation marks around the quoted material and include a footnote or other indication of the source of the quotation. This includes cut and paste from the Internet or other electronic sources;*
- 2. Changes words but copies the sentence structure of a source without giving credit to the original source, or closely paraphrases one or more paragraphs without acknowledgment of the source of the ideas, or uses graphs, figures, drawings, charts or other visual/audio materials without acknowledging the source or the permission of the author;*
- 3. Submits false or altered information in any academic exercise. This may include making up data for an experiment, altering data, citing nonexistent articles, contriving sources, etc.;*
- 4. Turns in all or part of assignment done by another student and claims it as their own;*
- 5. Uses a paper writing service, has another student write a paper, or uses a foreign language translation and submits it as their own original work.*

Disclaimer Subsequent changes can be made to any part of this syllabus when necessary. Any changes will be announced in class and on Moodle.